

PowerBI Topics:

1. Introducing MicroSoft Power BI Desktop

- Meet Power BI Desktop
- Downloading Power BI
- Important: Adjusting Settings
- Power BI Desktop Interface & Workflow
- Resources & Monthly Updates

2. Connecting & Shaping Data

- Power BI Front – End Vs Back End
- Types of Data Connectors
- The Power Query Editor
- Basic table Transformations
- Connecting to a Database
- Extracting Data from the web
- Data QA & Profiling Tools
- Text Tools
- Numerical Tools
- Date & time Tools
- Change Type with Locale
- Index & Conditional Columns
- Calculated Column best practices
- Grouping & Aggregating
- Pivoting & unpivoting
- Merging Queries
- Appending Queries
- Appending Files form a Folder
- Data Sources Settings

- Refreshing Queries

3. Creating a Data Model

- Data Modeling 101
- Database Normalization
- Fact & Dimension Tables
- Primary & foreign Keys
- Relationships Vs. Merged Tables
- Creating Table Relationships
- Managing & Editing Relationships
- Star & Snowflake Schemas
- Active & inactive Relationships
- Relationship Cardinality
- Connecting Multiple Fact tables
- Filter Context & Filter Flow
- Bi- Directional Filters & Ambiguity
- Hiding Fields from Report view
- Model Layouts
- Data Formats& Categories
- Creating Hierarchies

4. Calculated Fields with DAX

- Data Analysis Expression 101
- Dax vs Mlanguages
- Intro to DAX calculated Columns
- Intro to DAX Measure
- Implicit vs Explicit Measures
- Quick Measures
- Calculated Columns vs. Measures

- Dedicated Measure Tables
- Understanding Filter Context
- Step -by-Step DAX Measure Calculation
- DAX syntax & Operators
- Common DAX Function Categories
- Basic Math & Stats Functions
- Counting Functions
- Conditional & logical Functions
- The SWITCH Function
- Common Text Functions
- Basic Date & Time Functions
- Joining Data with RELATED
- The CALCULATE Function
- DAX Measure Totals
- The ALL Function
- The FILTER function
- Time Intelligence Patterns

5. Visualizing Data with Reports

- The 3 Key Questions
- Dashboard Design Framework
- Sketching the Dashboard Layout
- Adding Report Pages & objects
- Cards & multi-Row cards
- Building & formatting Charts
- Line Charts
- Trend Lines & Forecasts
- KPI Cards

- Bar & Donut Charts
- Basic Filtering Options
- Table & matrix Visuals
- Conditional Formatting
- TOP N Filtering
- Top N Text Cards
- Map Visuals
- Report Slicers
- Gauge Charts
- Advanced Conditional formatting
- Area Charts
- Drill Up & Drill Down
- Drillthrough Filters
- Editing Report Interactions
- Custom Navigation Buttons
- PRO TIP: Slicer Panels
- PRO TIP: Custom Tool Tips
- Importing Custom visuals
- Managing & viewing Roles
- Mobile Layouts
- Publishing Power BI Service